Food Microbiology

Complete lesson notes

Full content from the supplied presentation and two reference documents. Read down the left column, then down the right column. Source panels preserve diagrams, tables, labels and the original wording alongside the selectable text. Repeated content is retained.

Module 10: Foodborne Diseases

Slide 1 - Module 10. Foodborne Diseases

Module 10. Foodborne Diseases

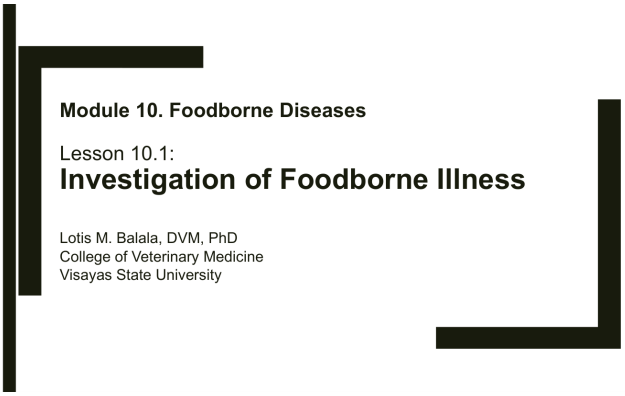
Lesson 10.1:

Investigation of Foodborne Illness

Lotis M. Balala, DVM, PhD

College of Veterinary Medicine

Visayas State University



Original source panel - slide 1

Slide 2 - Learning Outcomes

Learning Outcomes

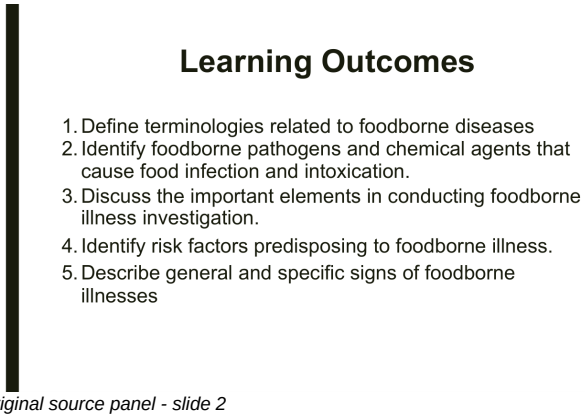
1.Define terminologies related to foodborne diseases

2.Identify foodborne pathogens and chemical agents that cause food infection and intoxication.

3.Discuss the important elements in conducting foodborne illness investigation.

4.Identify risk factors predisposing to foodborne illness.

5.Describe general and specific signs of foodborne illnesses



Original source panel - slide 2

Slide 3 - Foodborne Illness

Foodborne Illness

- illness/es acquired through consumption of food and water contaminated with pathogens or certain chemicals

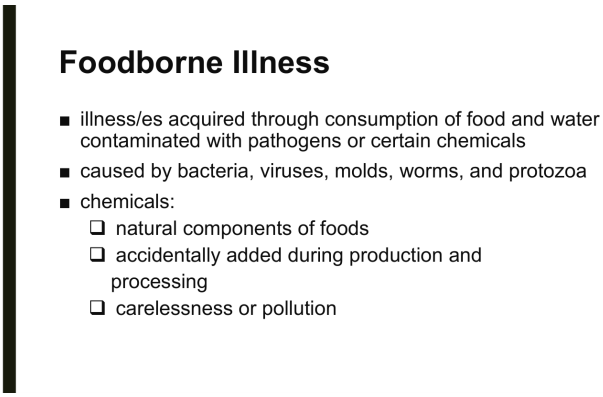
- caused by bacteria, viruses, molds, worms, and protozoa

- chemicals:

❑natural components of foods

❑accidentally added during production and processing

❑carelessness or pollution



Original source panel - slide 3

Slide 4 - ■ Allergens: nuts, milk, eggs, fish,

■ Allergens: nuts, milk, eggs, fish,

crustacean shellfish, tree nuts,

- The main causes of peanuts, wheat or soybeans,

foodborne illness are: can cause allergic reactions in

people with food allergies. ➢bacterial (66%)

➢chemical (26%)

➢viral (4%)

➢parasitic (4%)

■Food allergy is an

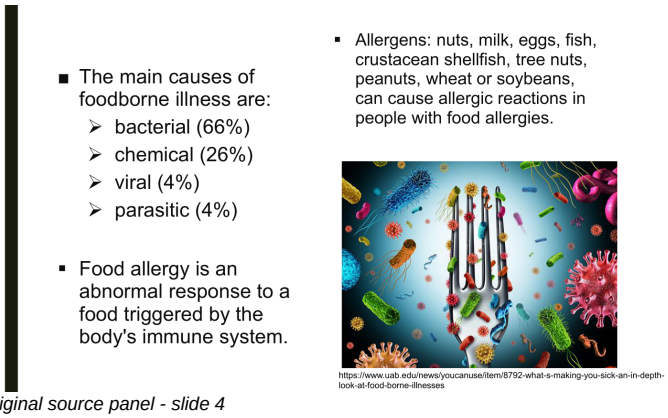
abnormal response to a

food triggered by the

body's immune system.

https://www.uab.edu/news/youcanuse/item/8792-what-s-making-you-sick-an-in-depth-

look-at-food-borne-illnesses



Original source panel - slide 4

Slide 5 - https://www.nestlenutrition-institute.org/

https://www.nestlenutrition-institute.org/



Original source panel - slide 5

Slide 6 - Classification of foodborne illness

Classification of foodborne illness

- Foodborne infection

– caused by the ingestion of food contaminated with viable pathogenic organisms that can establish infection

– after ingestion, the organism colonizes the digestive tract and multiplies in number, then spread to other parts of the body via blood circulation

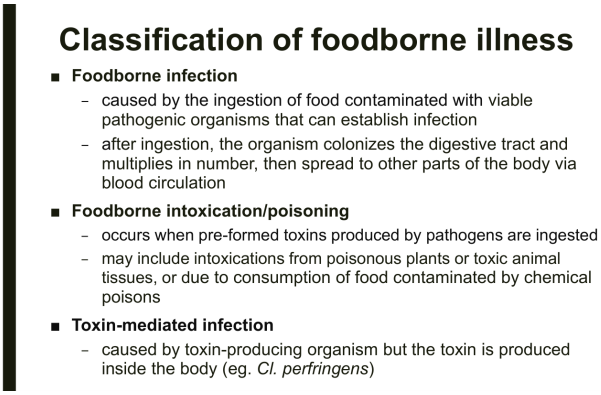
- Foodborne intoxication/poisoning

– occurs when pre-formed toxins produced by pathogens are ingested

– may include intoxications from poisonous plants or toxic animal tissues, or due to consumption of food contaminated by chemical poisons

- Toxin-mediated infection

– caused by toxin-producing organism but the toxin is produced inside the body (eg. Cl. perfringens)



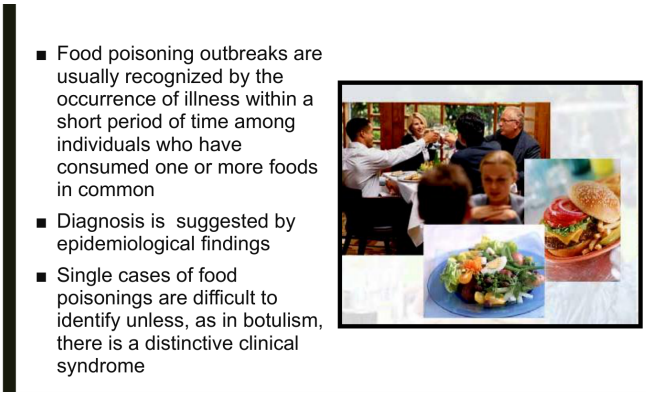
Original source panel - slide 6

Slide 7 - ■Food poisoning outbreaks are

- Food poisoning outbreaks are usually recognized by the occurrence of illness within a short period of time among individuals who have consumed one or more foods in common

- Diagnosis is suggested by epidemiological findings

- Single cases of food poisonings are difficult to identify unless, as in botulism, there is a distinctive clinical syndrome



Original source panel - slide 7

Slide 8 - Food intoxication Food infection

Food intoxication Food infection

- Bacillus cereus- Campylobacter jejuni

- Clostridium botulinum- Escherichia coli 0157:H7

- Clostridium perfringens- Hepatitis A virus (cafeteria germ)

- Listeria monocytogenes

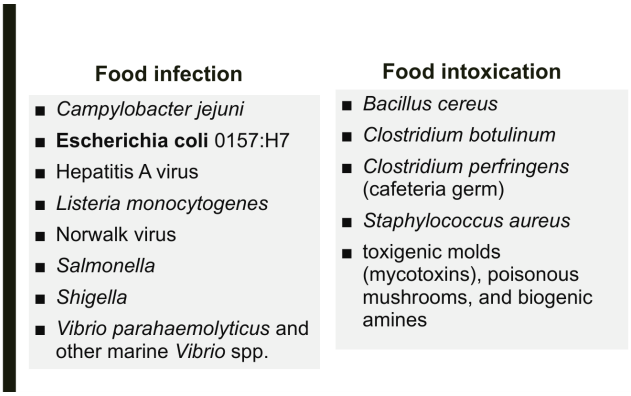
- Staphylococcus aureus

- Norwalk virus

- toxigenic molds

- Salmonella (mycotoxins), poisonous

- Shigella mushrooms, and biogenic amines- Vibrio parahaemolyticus and other marine Vibrio spp.



Original source panel - slide 8

Slide 9 - Investigation of Foodborne Illness

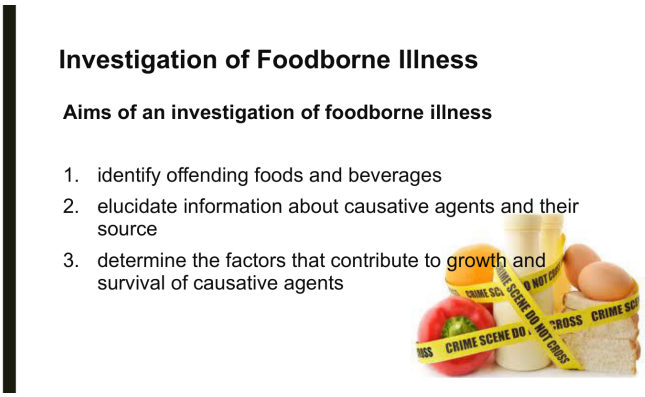
Investigation of Foodborne Illness

Aims of an investigation of foodborne illness

1. identify offending foods and beverages

2. elucidate information about causative agents and their source

3. determine the factors that contribute to growth and survival of causative agents



Original source panel - slide 9

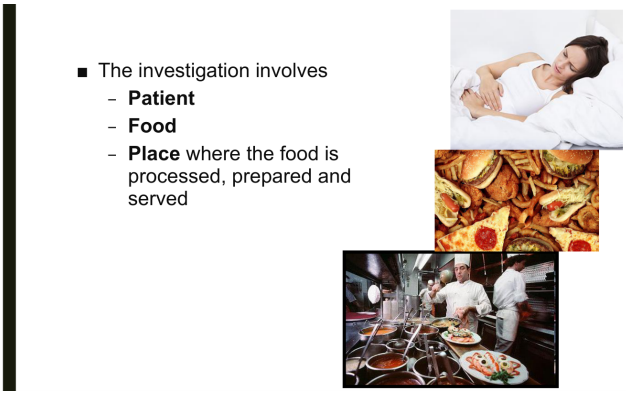
Slide 10 - ■The investigation involves

- The investigation involves

– Patient

– Food

– Place where the food is processed, prepared and served



Original source panel - slide 10

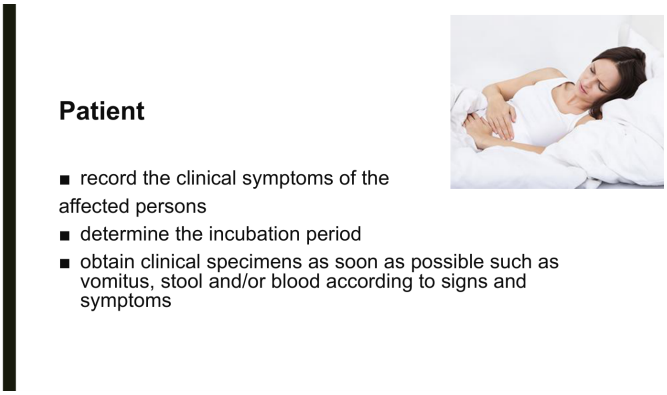
Slide 11 - Patient

Patient

- record the clinical symptoms of the affected persons

- determine the incubation period

- obtain clinical specimens as soon as possible such as vomitus, stool and/or blood according to signs and symptoms



Original source panel - slide 11

Slide 12 - Food

Food

- collect information about all meals and snacks eaten 72 hour before onset of illness

- collect leftovers from food that were eaten 72 hour before onset of illness or some ingredients that were used in foods and of any food available from an allegedly contaminated lot



Original source panel - slide 12

Slide 13 - Place

Place

1. Investigate the hygienic condition at the place where the suspect

foods were processed, prepared and served.

Try to identify the circumstances that permitted survival and growth

of food-borne pathogens such as:

i) inadequate refrigeration

ii) inadequate cooking and reheating

iii) holding foods at warm temperature several hours before serving

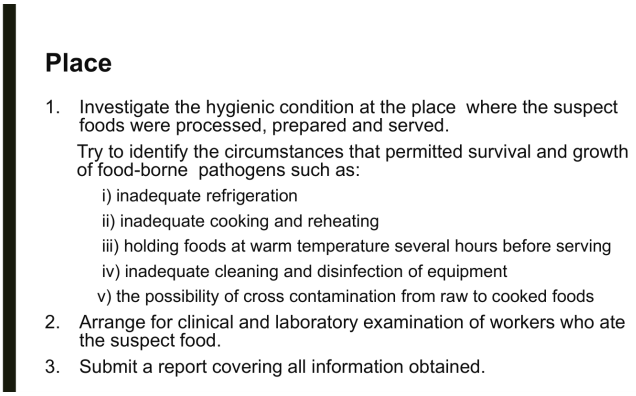
iv) inadequate cleaning and disinfection of equipment

v) the possibility of cross contamination from raw to cooked foods

2. Arrange for clinical and laboratory examination of workers who ate

the suspect food.

3. Submit a report covering all information obtained.



Original source panel - slide 13

Slide 14 - Summary of steps in the investigation of food-

Summary of steps in the investigation of food-

borne disease outbreak investigation

1. Verify the existence of an outbreak ƒ

■ Compare the current number of cases with the past

Note: -consider seasonal variations

2. Verify the diagnosis

■ Review clinical and laboratory findings

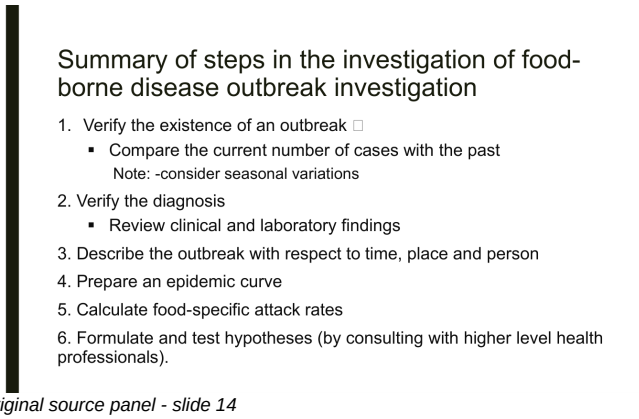
3. Describe the outbreak with respect to time, place and person

4. Prepare an epidemic curve

5. Calculate food-specific attack rates

6. Formulate and test hypotheses (by consulting with higher level health

professionals).



Original source panel - slide 14

Slide 15 - 7. Search for additional cases

7. Search for additional cases

8. Analyze the data

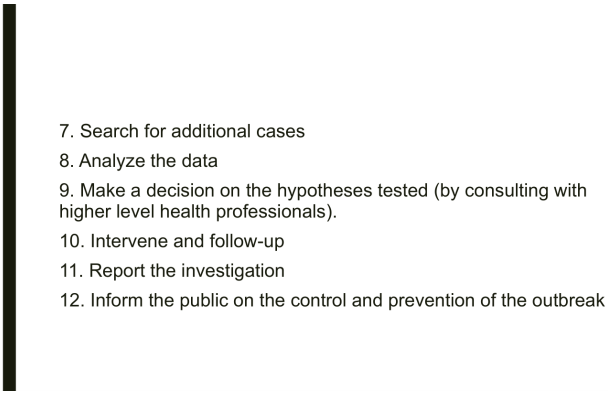
9. Make a decision on the hypotheses tested (by consulting with

higher level health professionals).

10. Intervene and follow-up

11. Report the investigation

12. Inform the public on the control and prevention of the outbreak



Original source panel - slide 15

Slide 16 - Top 5 Foodborne Illness Risk Factors

Top 5 Foodborne Illness Risk Factors

1. Poor Personal Hygiene

■Poor personal hygiene practices serve as the leading cause of

foodborne illnesses. Food establishments must promote a culture

of food safety by developing an employee illness policy, proper

handwashing procedure, and a no barehand contact policy with

ready to eat foods

2. Improper Holding Temperatures

■Cold foods shall maintain ≤41°F and hot foods shall maintain

≥135°F. If foods stay in between those temperatures for any period,

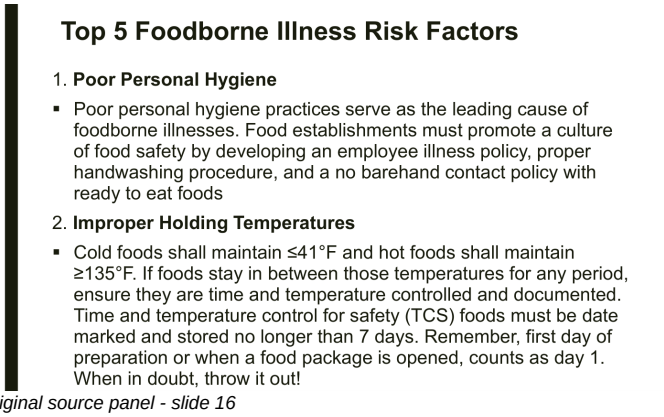
ensure they are time and temperature controlled and documented.

Time and temperature control for safety (TCS) foods must be date

marked and stored no longer than 7 days. Remember, first day of

preparation or when a food package is opened, counts as day 1.

When in doubt, throw it out!



Original source panel - slide 16

Slide 17 - 3. Improper Cooking Temperatures

3. Improper Cooking Temperatures

■Ensure metal stem thermometers are conveniently stored

and accessible for employees to monitor final cooking

temperatures.

4. Food from Unsafe Sources

■All foods distributed in licensed food establishments must be obtained from approved sources that comply with applicable laws and regulations. All food shall be inspected upon delivery to ensure proper temperatures, it's condition and overall sanitation.

3. **Improper Cooking Temperatures**
- Ensure metal stem thermometers are conveniently stored and accessible for employees to monitor final cooking temperatures.

4. **Food from Unsafe Sources**
- All foods distributed in licensed food establishments must be obtained from approved sources that comply with applicable laws and regulations. All food shall be inspected upon delivery to ensure proper temperatures, it's condition and overall sanitation.

Original source panel - slide 17

Slide 18 - 5. Contaminated Equipment/Cross-Contamination

5. Contaminated Equipment/Cross-Contamination

■Cross contamination can occur from a variety of sources, including chemical and raw foods.

■To prevent the spread of harmful pathogens

■all equipment and utensils shall be properly cleaned and sanitized at least once every 4 hours

■ensure sanitizer solutions are monitored using their appropriate test strips

■low sanitizer concentrations will fail in effectively removing pathogens from surfaces, while high concentrations will leave a toxic residue

5. **Contaminated Equipment/Cross-Contamination**
- Cross contamination can occur from a variety of sources, including chemical and raw foods.
- To prevent the spread of harmful pathogens
  - all equipment and utensils shall be properly cleaned and sanitized at least once every 4 hours
  - ensure sanitizer solutions are monitored using their appropriate test strips
  - low sanitizer concentrations will fail in effectively removing pathogens from surfaces, while high concentrations will leave a toxic residue

Original source panel - slide 18

Slide 19 - Effects of food poisoning

Effects of food poisoning

- develop symptoms after ingesting a pathogen

- others never know that they are suffering from food-borne illness

- High risk:

1. compromised immune systems
2. elderly
3. very young

Effects of food poisoning
- develop symptoms after ingesting a pathogen
- others never know that they are suffering from food-borne illness
- High risk:
  1. compromised immune systems
  2. elderly
  3. very young



Original source panel - slide 19

Slide 20 - ■Symptoms of foodborne illnesses depend on the cause

- Symptoms of foodborne illnesses depend on the cause

- Common symptoms of many foodborne illnesses include vomiting, diarrhea or bloody diarrhea, abdominal pain, fever, and chills

- Foodborne illnesses may lead to dehydration, hemolytic uremic syndrome (HUS), and other complications

- Acute foodborne illnesses may also lead to chronic— or long lasting—health problems

- Treatment: replacing lost fluids and electrolytes to prevent dehydration.

- Foodborne illnesses can be prevented by properly storing, cooking, cleaning, and handling foods.

- Symptoms of foodborne illnesses depend on the cause
- Common symptoms of many foodborne illnesses include **vomiting, diarrhea or bloody diarrhea, abdominal pain, fever, and chills**
- Foodborne illnesses may lead to dehydration, hemolytic uremic syndrome (HUS), and other complications
- Acute foodborne illnesses may also lead to chronic— or long lasting—health problems
- Treatment: replacing lost fluids and electrolytes to prevent dehydration.
- Foodborne illnesses can be prevented by properly storing, cooking, cleaning, and handling foods.

Original source panel - slide 20

Slide 21 - Lesson 10.2:

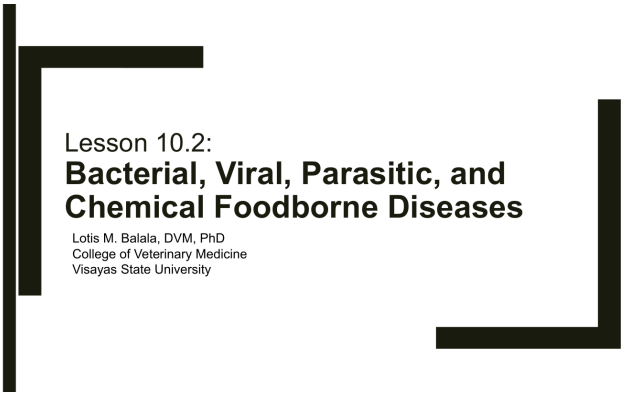
Lesson 10.2:

Bacterial, Viral, Parasitic, and Chemical Foodborne Diseases

Lotis M. Balala, DVM, PhD

College of Veterinary Medicine

Visayas State University



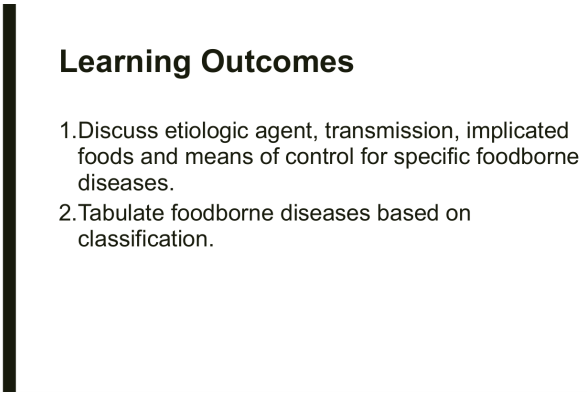
Original source panel - slide 21

Slide 22 - Learning Outcomes

Learning Outcomes

1.Discuss etiologic agent, transmission, implicated foods and means of control for specific foodborne diseases.

2.Tabulate foodborne diseases based on classification.

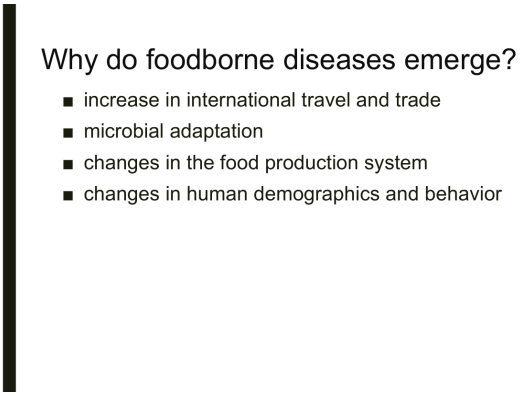


Original source panel - slide 22

Slide 23 - Why do foodborne diseases emerge?

Why do foodborne diseases emerge?

- increase in international travel and trade
- microbial adaptation
- changes in the food production system
- changes in human demographics and behavior



Original source panel - slide 23

Slide 24 - Other contributing factors

Other contributing factors

- The globalization of the food supply

– Importation of contaminated raspberries from South America led

to a large outbreak of cyclosporiasis in North America in 1996-97

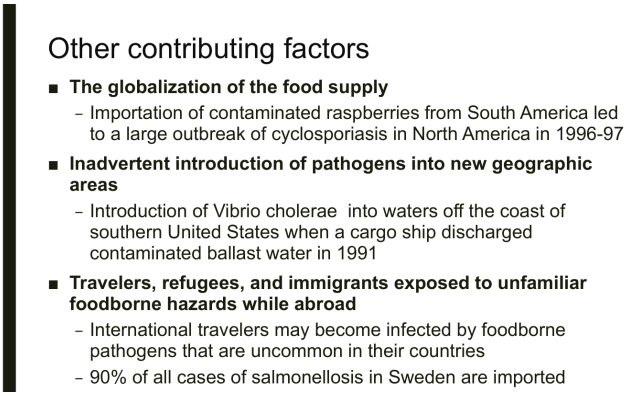
- Inadvertent introduction of pathogens into new geographic areas

– Introduction of Vibrio cholerae into waters off the coast of southern United States when a cargo ship discharged contaminated ballast water in 1991

- Travelers, refugees, and immigrants exposed to unfamiliar foodborne hazards while abroad

– International travelers may become infected by foodborne pathogens that are uncommon in their countries

– 90% of all cases of salmonellosis in Sweden are imported



Original source panel - slide 24

Slide 25 - ■Changes in microorganisms

- Changes in microorganisms

– evolution of new pathogens

– development of new virulent strains in old pathogens

– development of antibiotic resistance pathogens

– ability to survive in adverse environmental conditions

- Change in the human population

– population of highly susceptible persons is expanding world-wide

- ageing, malnutrition, HIV infections and other underlying medical conditions

- reduced immunity due to poor nutritional status render people more susceptible to foodborne infections

- **Changes in microorganisms**
  - evolution of new pathogens
  - development of new virulent strains in old pathogens
  - development of antibiotic resistance pathogens
  - ability to survive in adverse environmental conditions
- **Change in the human population**
  - population of highly susceptible persons is expanding world-wide
    - ageing, malnutrition, HIV infections and other underlying medical conditions
    - reduced immunity due to poor nutritional status render people more susceptible to foodborne infections

Original source panel - slide 25

Slide 26 - ■Changes in lifestyle

- Changes in lifestyle

– Greater numbers of people go out and eat meals prepared in restaurants, canteens, fast food outlets, and by street food vendors

– Boom in food service establishments is not matched by effective food safety education and control

– Unhygienic preparation of food provides ample opportunities for contamination, growth, or survival of foodborne pathogens

- **Changes in lifestyle**
  - Greater numbers of people go out and eat meals prepared in restaurants, canteens, fast food outlets, and by street food vendors
  - Boom in food service establishments is not matched by effective food safety education and control
  - Unhygienic preparation of food provides ample opportunities for contamination, growth, or survival of foodborne pathogens

Original source panel - slide 26

Slide 27 - Visual content

DISEASE	PATHOGEN	INCUBATION PERIOD/MEAN	DURATION OF ILLNESS	SYMPTOMS	FOODS IMPLICATED	TYPE OF ILLNESS	RESERVOIR
Salmonellosis	*Salmonella* (facultative) BACTERIA	8-72 hours 18-36 hours	2-3 days	Abdominal pain, headache, fever, nausea, diarrhea, chills, cramps	Poultry, eggs, meat, meat products, milk, smoked fish, protein foods, juice	Infection	Domestic and wild animals, humans
Shigellosis	*Shigella* (facultative) BACTERIA	1-7 days 1-3 days	Indefinite	Diarrhea, fever, chills, cramps, lassitude, nausea, dehydration	Moist, mixed foods, milk, potato, tuna, bean, and macaroni salads, apple cider	Infection	Human feces, flies
Listeriosis	*Listeria monocytogenes* BACTERIA	2 days – 3 weeks 4-21 days	Indefinite, high fatality in the immunocompromised	Nausea, vomiting, chills, headache, fever, backache, meningitis	Unpasteurized dairy foods, vegetables, meat, poultry, seafood, raw and smoked fish, ready-to-eat foods	Infection	Domestic and wild animals, humans, soil
Staphylococcus	*Staphylococcus aureus* BACTERIA	1-6 hours 2-4 hours	1-2 days	Nausea, cramps, vomiting, dehydration	Reheated foods, ham, poultry, dairy foods, custards, potato salad, cream-filled foods	Intoxication	Humans (skin, throat, nose), animals
Clostridium perfringens	*Clostridium perfringens* BACTERIA	8-22 hours 10 hours	24 hours	Abdominal pain, diarrhea, gas	Meat, meat products, gravy, thick foods that are held at a low temp or cooled slowly	Infection/ Intoxication	Human intestinal tract, animals, soil
Bacillus cereus	*Bacillus cereus* BACTERIA	½ - 5 hours; 8-16 hours ½ -5 hrs; 12 hrs	6-24 hours; 12 hours	Nausea and vomiting; Diarrhea, abdominal cramps	Rice, custards, seasonings, dry food mixes, spices, salads, casseroles, milk, sauces, meats	Intoxication	Soil, dust

Original source panel - slide 27

Slide 28 - Visual content

					meats		
Campylobacter	*Campylobacter jejuni* BACTERIA	2-5 days	5-7 days 3-5 days	Diarrhea which may be watery or bloody, fever, nausea, headache	Chicken, unpasteurized milk, water	Infection	Domestic and wild animals (intestinal tract)
E. coli 0157:H7	*Escherichia coli* BACTERIA	2-5 days	5-10 days 2-5 days	Diarrhea (may be bloody), abdominal pain, nausea, vomiting, fever, HUS	Undercooked ground beef, unported cheeses, unpasteurized milk or juice, cider, alfalfa sprouts	Infection/ Intoxication	Human intestinal tract, animals (especially cattle)
Botulism	*Cloatridium botulinum* BACTERIA	12-36 hours, 72 or more hours	Several days – year 18-36 hours	Vertigo, inability to swallow, respiratory paralysis, visual disturbances	Improperly processed low-acid canned goods, garlic in oil, grilled onions, meat, stew, baked potatoes	Intoxication	Soil, water
Viral Gastroenteritis	Snow Mountain, calicivirus virus	24-48 hours	24-60 hours 36 hours	Diarrhea, vomiting, nausea, cramps, headache, muscle aches, tiredness, slight fever	Water, ready-to-eat foods (salad, sandwiches, bread) shellfish	N/A	Human intestinal tract
Hepatitis A	Hepatitis A VIRUS	7-50 days	Several weeks to months 25-30 days	Fever, fatigue, stomach pain, nausea, dark urine, jaundice	Ready-to-eat foods, fruit and juice, milk products, shellfish, salads, vegetables, sandwiches, water	N/A	Infected people
Norovirusn infection	Noro Virus	12-96 hours	1-2 days	Low-grade fever, chills, headache, muscle ache, and fatigue	Ready-to-eat foods (salad, sandwiches, bread) shellfish	Infection	Human intestinal tract

Original source panel - slide 28

Slide 29 - Also read PDF copy of:

Also read PDF copy of:

Also read PDF copy of:

Foodborne Illness Quick Reference

Page 1	General Information
Page 2	Anisakiasis
Page 3	Bacillus cereus
Page 4	Botulism
Page 5	Campylobacteriosis
Page 6	Ciguatera Fish Poisoning
Page 7	Clostridium Perfringens
Page 8	E. coli O157:H7
Page 9	Food Allergens
Page 10	Hepatitis A

Original source panel - slide 29

Lesson 9.2: Microbial Spoilage of Canned Food

Page 1 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

Lesson 9.2

Microbial Spoilage of Canned Food

vsu.edu.ph



Original source panel - page 1

Page 2 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

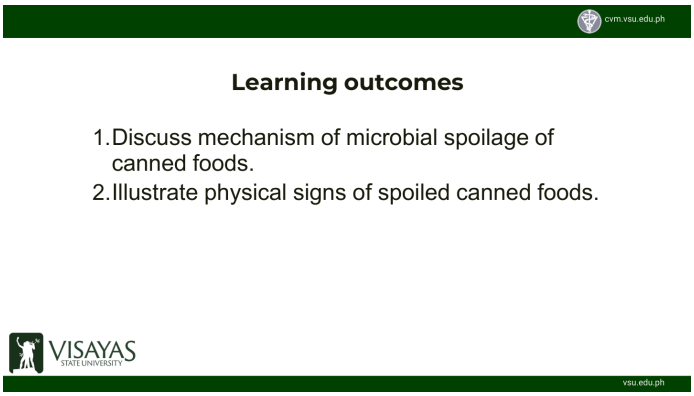
Learning outcomes

1.Discuss mechanism of microbial spoilage of

canned foods.

2.Illustrate physical signs of spoiled canned foods.

vsu.edu.ph



Original source panel - page 2

Page 3 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

Factors Affecting Microbial Spoilage

- Spoilage of canned goods may be due to

a. Microbes

b. Chemicals

c. Deleterious influences such as rust and damage

- Canned goods are classified as spoiled when the food has undergone

a deleterious change or when the condition of the container renders

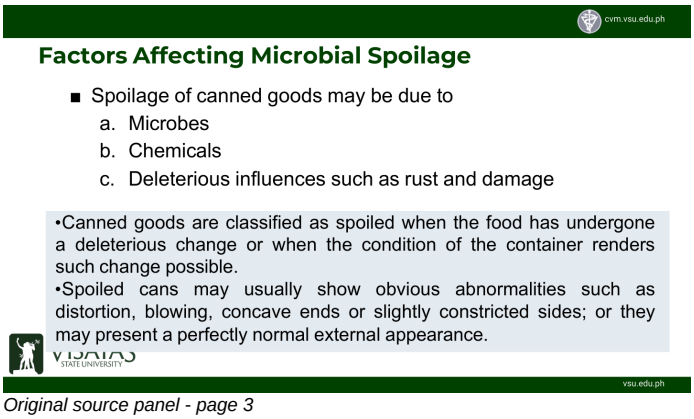
such change possible.

- Spoiled cans may usually show obvious abnormalities such as

distortion, blowing, concave ends or slightly constricted sides; or they

may present a perfectly normal external appearance.

vsu.edu.ph



Original source panel - page 3

Page 4 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

Microbial spoilage

- Type of microorganisms

- bacteria of decomposing or fermenting type are the most important

cause while the spore forming bacteria are the most resistant

- three main types of spore forming organism, which can resist normal

processing and may cause spoilage in canned foods:

a. Gas producing anaerobic organisms growing at an optimum

temperature of 55°C .

b. Gas producing anaerobic and aerobic organisms with an

optimum temperature growth of 37°C.

§ Both of the above types are associated with the form of spoilage

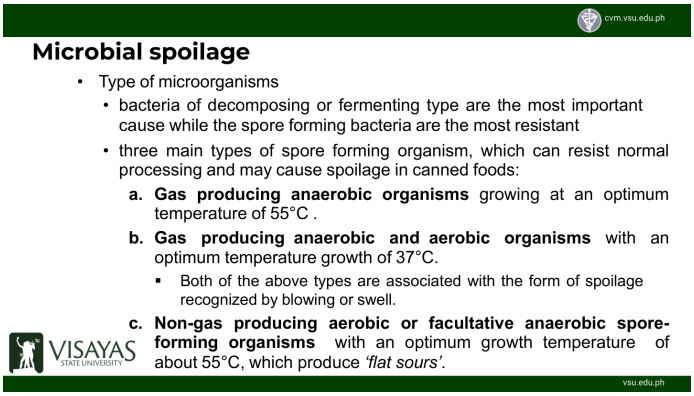
recognized by blowing or swell.

c. Non-gas producing aerobic or facultative anaerobic spore-

forming organisms with an optimum growth temperature of

about 55°C, which produce ‘flat sours’.

vsu.edu.ph



Original source panel - page 4

Page 5 - The factors determining microbial, unsoundness in cvm.vsu.edu.ph

The factors determining microbial, unsoundness in cvm.vsu.edu.ph

canned goods are

- The number of microorganisms present

- Processing is not a substitute for cleanliness; a small number of

bacteria are likely to be destroyed more easily by processing than are

a large numbers.

- Efficiency of processing

- Insufficient processing must be regarded as a cause of unsoundness

of canned foods

- Access of the air to the can

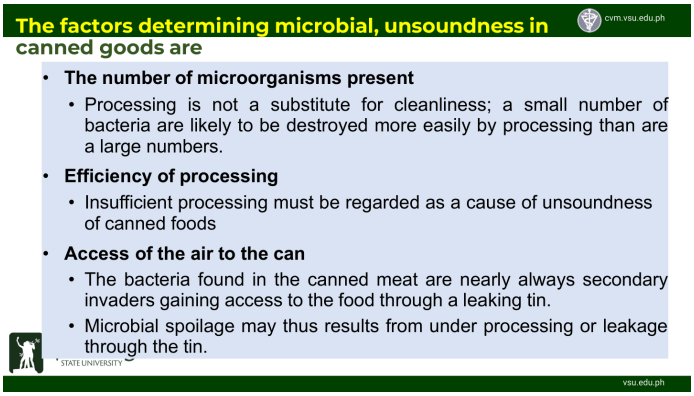
- The bacteria found in the canned meat are nearly always secondary

invaders gaining access to the food through a leaking tin.

- Microbial spoilage may thus results from under processing or leakage

through the tin.

vsu.edu.ph



Original source panel - page 5

Page 6 - cvm.vsu.edu.phFlat Souring

cvm.vsu.edu.phFlat Souring

- manifested by the presence of a sour odor of foodstuffs, but

without the can becoming blown.

- Canned foods susceptible to this deleterious change are those

containing sugar or starches, and meat products such as

sausages or paste containing cereals.

- True flat sours are commonly caused by the growth of

thermophilic organisms (B. coagulans, B. stearothermophilus, B.

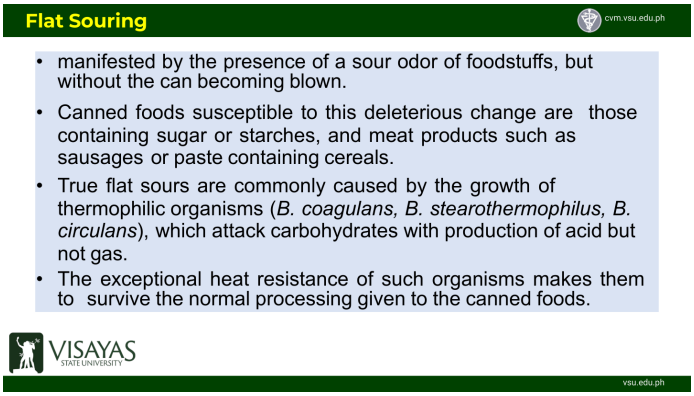
circulans), which attack carbohydrates with production of acid but

not gas.

- The exceptional heat resistance of such organisms makes them

to survive the normal processing given to the canned foods.

vsu.edu.ph



Original source panel - page 6

Page 7 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

- Sourness in canned foods may also arise due to leaking

cans or it may have developed in the food before

processing.

- In corned beef, the presence of souring without gas

production may be due to under-processing or leaking seams,

but the development of pre-processing storage is also likely to be

the cause.

- flat sours of canned foods due to thermophilic spore forming

organisms can not be detected until the can is opened and its

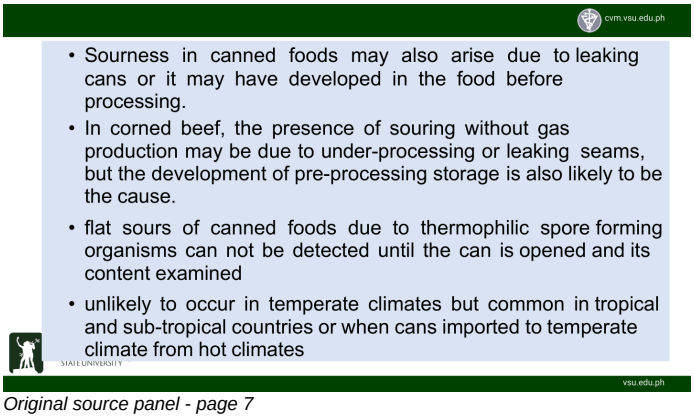
content examined

- unlikely to occur in temperate climates but common in tropical

and sub-tropical countries or when cans imported to temperate

climate from hot climates

vsu.edu.ph



Original source panel - page 7

Page 8 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

Chemical spoilage - Hydrogen swell

- most important problems encountered in the canning

industry owing to spoilage on account of chemical

changes such as:

- Hydrogen swelling

- Purple staining

- Rust or damage

Hydrogen Swell

- happens independent of microbial spoilage, and is associated with the

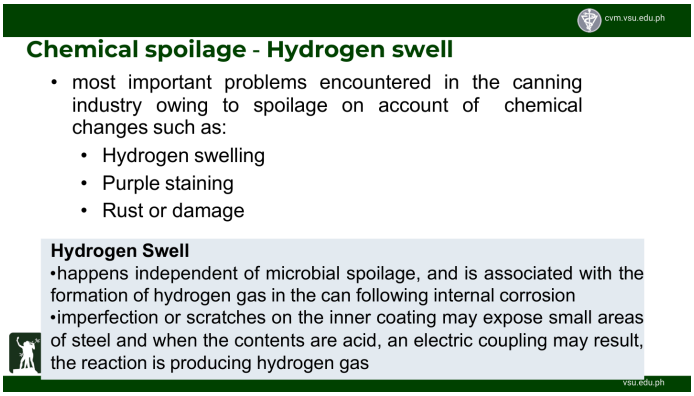
formation of hydrogen gas in the can following internal corrosion

- imperfection or scratches on the inner coating may expose small areas

of steel and when the contents are acid, an electric coupling may result,

the reaction is producing hydrogen gas

vsu.edu.ph



Original source panel - page 8

Page 9 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

- Can affected with hydrogen swell may show varying degrees of

bulging from flipping to blowing and if the tin being punctured,

there is emission of hydrogen gas, whichis odorless and burns

on the application of a flame

- This condition is chiefly associated with food containing organic

acids particularly fruits such as plums, cherries, raspberries,

etc.

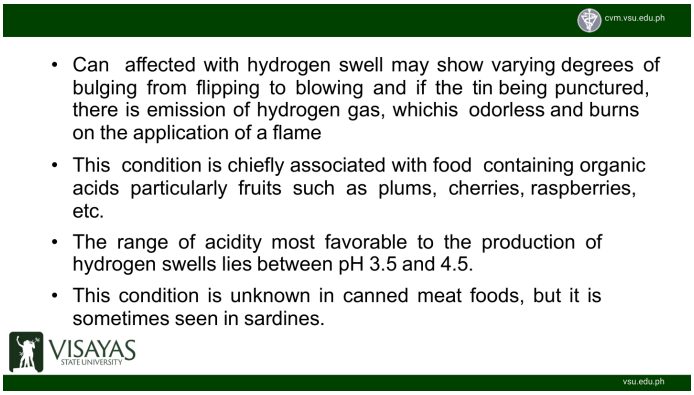
- The range of acidity most favorable to the production of

hydrogen swells lies between pH 3.5 and 4.5.

- This condition is unknown in canned meat foods, but it is

sometimes seen in sardines.

vsu.edu.ph



Original source panel - page 9

Page 10 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

Judgement

- Though the content of the can in hydrogen swell may be

quite harmless, the routine methods employed in the

examination of canned goods render it impossible to

distinguish between the tins which are blown due to hydrogen

swell and those which are blown as a result of deleterious

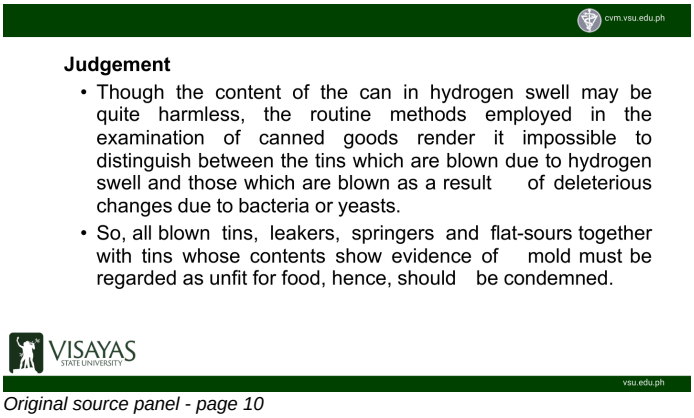
changes due to bacteria or yeasts.

- So, all blown tins, leakers, springers and flat-sours together

with tins whose contents show evidence of mold must be

regarded as unfit for food, hence, should be condemned.

vsu.edu.ph



Original source panel - page 10

Page 11 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

Sulphiding

- Purple staining on the inner surface of the cans in which sulphur-

containing foods are packed may occur with all fish and meat products,

especially, liver, kidneys and tongue.

- due to the breaking down of sulphur containing proteins at a high

processing temperature with a result that, hydrogen sulphide is liberated

and a thin layer of tin sulphide is formed on the inside of the can

- This discoloration, which does not involves foodstuff itself and varies from a

light pink to a dark purple but a blackening of both inside of the can and the

surface of the foodstuffs may accompany it.

- These latter changes are due to attack of hydrogen sulphide on the steel

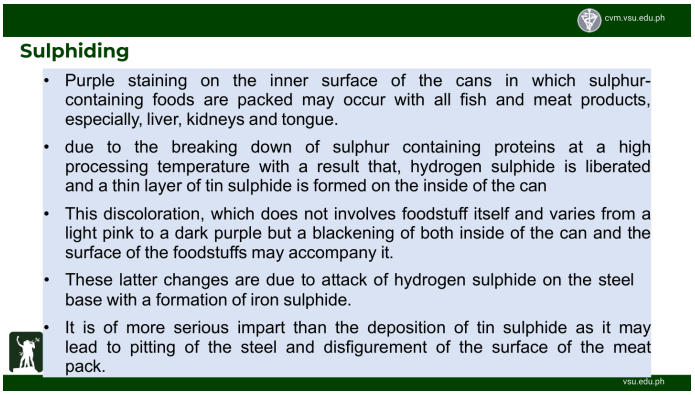
base with a formation of iron sulphide.

- It is of more serious impart than the deposition of tin sulphide as it may

lead to pitting of the steel and disfigurement of the surface of the meat

pack.

vsu.edu.ph



Original source panel - page 11

Page 12 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

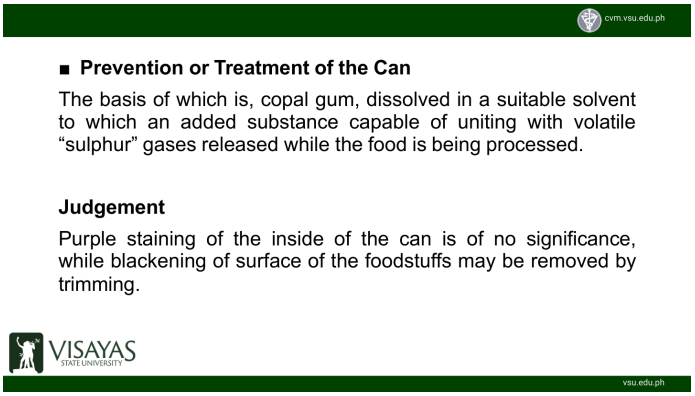
- Prevention or Treatment of the Can

The basis of which is, copal gum, dissolved in a suitable solvent to which an added substance capable of uniting with volatile “sulphur” gases released while the food is being processed.

Judgement

Purple staining of the inside of the can is of no significance, while blackening of surface of the foodstuffs may be removed by trimming.

vsu.edu.ph



Original source panel - page 12

Page 13 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

Rust or Damage

- external rust requires careful consideration and its conditions beneath can labels should be checked, esp. that adhesive contains hygroscopic substances

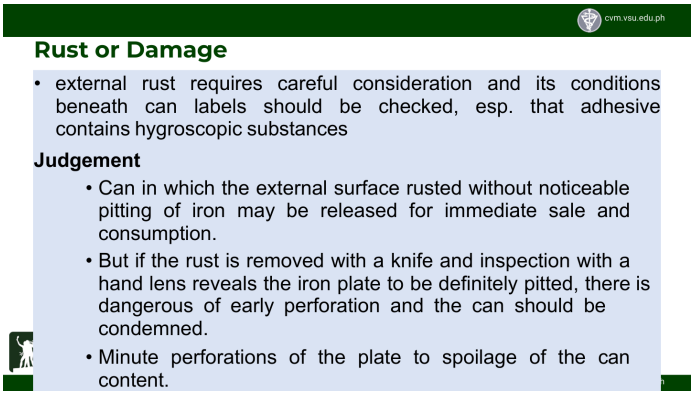
Judgement

- Can in which the external surface rusted without noticeable pitting of iron may be released for immediate sale and consumption.

- But if the rust is removed with a knife and inspection with a hand lens reveals the iron plate to be definitely pitted, there is dangerous of early perforation and the can should be condemned.

- Minute perforations of the plate to spoilage of the can

vsu.edu.ph content.



Original source panel - page 13

Page 14 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

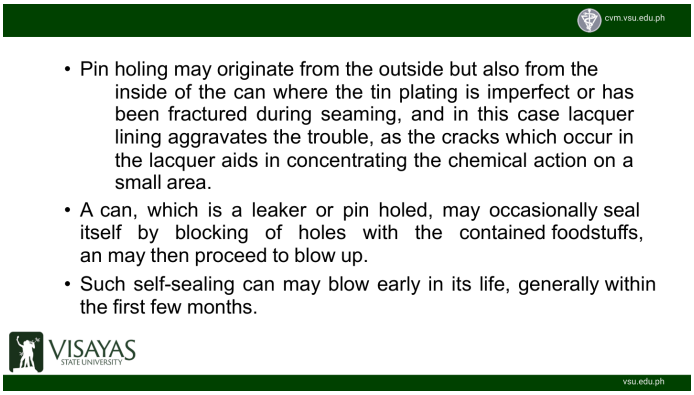
- Pin holing may originate from the outside but also from the inside of the can where the tin plating is imperfect or has been fractured during seaming, and in this case lacquer lining aggravates the trouble, as the cracks which occur in the lacquer aids in concentrating the chemical action on a small area.

- A can, which is a leaker or pin holed, may occasionally seal itself by blocking of holes with the contained foodstuffs,

an may then proceed to blow up.

- Such self-sealing can may blow early in its life, generally within the first few months.

vsu.edu.ph



Original source panel - page 14

Page 15 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

The various distortions commonly encountered are

- Swell or blower

- A can with it ends bulged by positive internal pressure due to gas

generated by microbial or chemical activity is termed as swell or

blower.

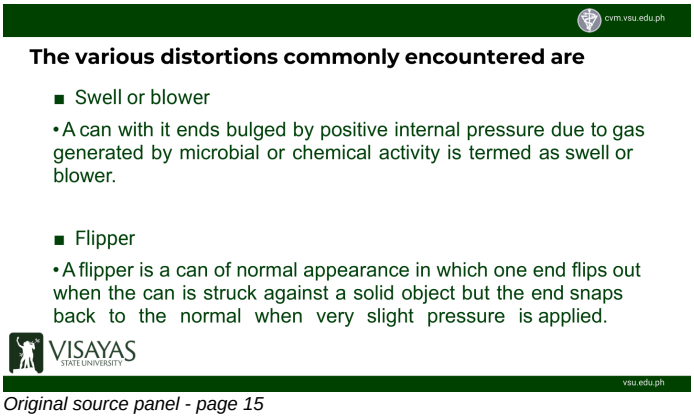
- Flipper

- A flipper is a can of normal appearance in which one end flips out

when the can is struck against a solid object but the end snaps

back to the normal when very slight pressure is applied.

vsu.edu.ph



Original source panel - page 15

Page 16 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

- Unfilled cans

- Where unfilled cans are stored and allowed to rust internally

before being filled, the can edges may become rusted, with

the result that during processing a chemical action may take

place between the rust and meat juices and give rise to an

unsightly grey precipitate of iron phosphate in the meat jelly.

- Damage

- Considerable significance should be attached to cans

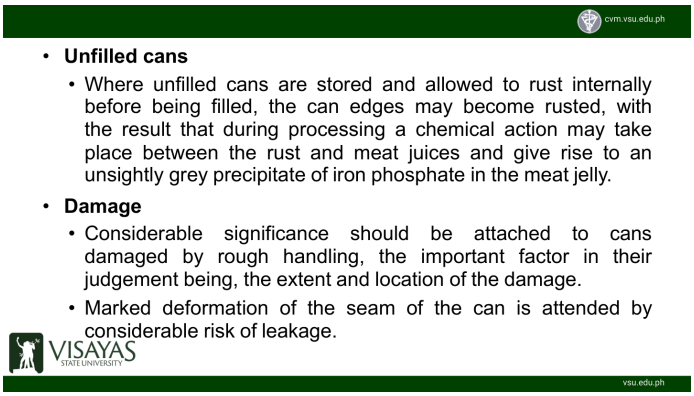
damaged by rough handling, the important factor in their

judgement being, the extent and location of the damage.

- Marked deformation of the seam of the can is attended by

considerable risk of leakage.

vsu.edu.ph



Original source panel - page 16

Page 17 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

Judgement

- Can showing seam damage should be condemned.

- Slight indentations on the can body are permissible, but severe

dents on the body may cause seam distention with danger of

leakage and such can should be rejected.

- Any can having a dent at one end should also be rejected for it is

possible to reduce a springer to normal, at any rate temporarily, by

hitting it upon the corner of a box.

- Nail holes in cans cause during closing of packaging cases may

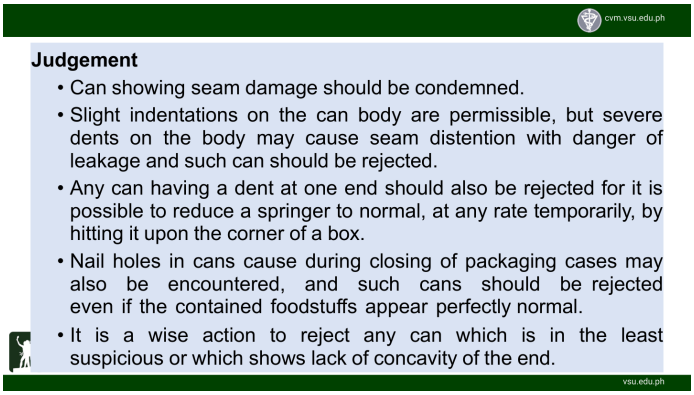
also be encountered, and such cans should be rejected

even if the contained foodstuffs appear perfectly normal.

- It is a wise action to reject any can which is in the least

suspicious or which shows lack of concavity of the end.

vsu.edu.ph



Original source panel - page 17

Page 18 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

- Springer

A springer is the term used to describe a can in which one end is bulged but can be

forced back into normal position, where upon the opposite end bulges.

- Leaker

- A leaker is a can containing a perforation from any cause whereby

atmospheric air may enter the can or its contents escape.

- May be detected by disappearance of vacuum from the sides and ends of the

cans and the bubbles that appear from the can when held under water and

squeezed.

- The commonest leaking is at the seams, and may sometimes be

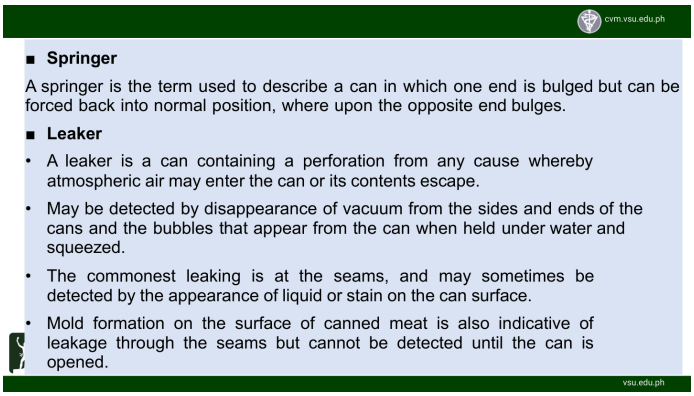
detected by the appearance of liquid or stain on the can surface.

- Mold formation on the surface of canned meat is also indicative of

leakage through the seams but cannot be detected until the can is

opened.

vsu.edu.ph



Original source panel - page 18

Page 19 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

- examination of the quality of containers is important to ensure that

there are no damaged, rusty, blown, etc. cans

- for non-shelf-stable products, five containers are examined

visually and their contents examined microbiologically

- Both aerobic and anaerobic techniques are undertaken,

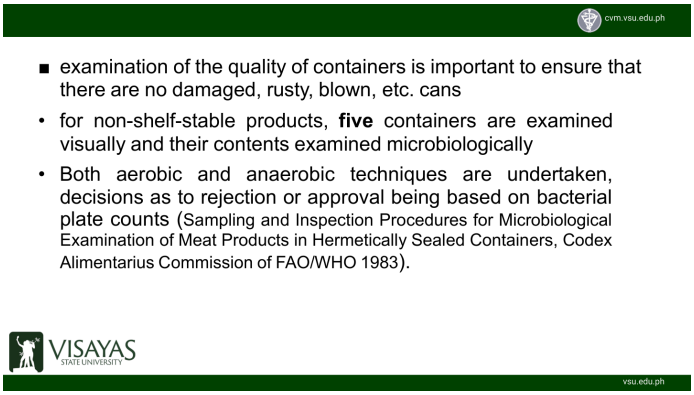
decisions as to rejection or approval being based on bacterial

plate counts (Sampling and Inspection Procedures for Microbiological

Examination of Meat Products in Hermetically Sealed Containers, Codex

Alimentarius Commission of FAO/WHO 1983).

vsu.edu.ph



Original source panel - page 19

Page 20 - cvm.vsu.edu.ph

cvm.vsu.edu.ph

vsu.edu.ph



Original source panel - page 20

Foodborne Illness Reference

Page 1 - New Mexico Environment Department

New Mexico Environment Department

Foodborne Illness Quick Reference

Page 1 General Information

Page 2 Anisakiasis

Page 3 Bacillus cereus

Page 4 Botulism

Page 5 Campylobacteriosis

Page 6 Ciguatera Fish Poisoning

Page 7 Clostridium Perfringens

Page 8 E. coli O157:H7

Page 9 Food Allergens

Page 10 Hepatitis A

Page 11 Listeriosis

Page 12 Norovirus

Page 13 Salmonellosis

Page 14 Scombroid Poisoning

Page 15 Shigellosis

Page 16 Staphylococcus

Page 17 Vibrio spp.

August 20, 2004



New Mexico Environment Department

Foodborne Illness Quick Reference

Page 1	General Information
Page 2	Anisakiasis
Page 3	Bacillus cereus
Page 4	Botulism
Page 5	Campylobacteriosis
Page 6	Ciguatera Fish Poisoning
Page 7	Clostridium Perfringens
Page 8	E. coli O157:H7
Page 9	Food Allergens
Page 10	Hepatitis A
Page 11	Listeriosis
Page 12	Norovirus
Page 13	Salmonellosis
Page 14	Scombroid Poisoning
Page 15	Shigellosis
Page 16	Staphylococcus
Page 17	Vibrio spp.

August 20, 2004

Original source panel - page 1

Page 2 - General Information

General Information

Classification of FBI's

- Foodborne Infection-

o After ingestion the organism burrows into the digestive tract and

multiplies in number.

o Can spread to other parts of the body through the blood stream.

o Bacteria, viruses, and parasites can cause FB infections.

o Example: Salmonellosis.

- Intoxication-

o Caused when a living organism multiplies in or on a food and then

produces a chemical waste or toxin.

o After consumption the toxin causes illness (typically called food

poisoning).

o An intoxication may also be caused by man-made chemicals (bleach)

o Examples: Clostridium botulinum and Staphylococcus aureus.

- Toxin-mediated infection-

o Caused by consuming a living organism, which then produces a toxin that

causes illness.

o Differs from an intoxication because the toxin is produces inside the body.

o Example: Clostridium perfringens.

Bacteria

- All bacteria exist in a vegetative form

o Vegetative cells grow, reproduce, and produce wastes just like other

living organisms.

- Some bacteria have the ability to form spores

o Spores help bacteria survive when the environment is too hot, cold, dry,

acidic, or when there is not enough food.

o Spores are not able to grow reproduce.

Vegetative Cells Spores

(Optimal Conditions) (Stress Conditions)

Reproduce Yes No

Grow Yes No

Produce Toxin Yes No

Resistant to Stress No Yes

Harmful if Eaten Yes No

General Information

Classification of FBI's

- Foodborne Infection-
  - After ingestion the organism burrows into the digestive tract and multiplies in number.
  - Can spread to other parts of the body through the blood stream.
  - Bacteria, viruses, and parasites can cause FB infections.
  - Example: Salmonellosis.
- Intoxication-
  - Caused when a living organism multiplies in or on a food and then produces a chemical waste or toxin.
  - After consumption the toxin causes illness (typically called food poisoning).
  - An intoxication may also be caused by man-made chemicals (bleach)
  - Examples: Clostridium botulinum and Staphylococcus aureus.
- Toxin-mediated infection-
  - Caused by consuming a living organism, which then produces a toxin that causes illness.
  - Differs from an intoxication because the toxin is produces inside the body.
  - Example: Clostridium perfringens.

Bacteria

- All bacteria exist in a vegetative form
  - **Vegetative cells** grow, reproduce, and produce wastes just like other living organisms.
- Some bacteria have the ability to form spores
  - **Spores** help bacteria survive when the environment is too hot, cold, dry, acidic, or when there is not enough food.
  - **Spores** are not able to grow reproduce.

	Vegetative Cells (Optimal Conditions)	Spores (Stress Conditions)
Reproduce	Yes	No
Grow	Yes	No
Produce Toxin	Yes	No
Resistant to Stress	No	Yes
Harmful if Eaten	Yes	No

Original source panel - page 2

Page 3 - Temperature Conditions

Temperature Conditions

- Psychrophilic bacteria grow within the range of 32°F to 70°F

o Most are spoilage organisms, but some cause illness

- Mesophilic bacteria grow within the range of 70°F to 110°F

- Thermophilic bacteria grow above 110°F

o All thermophilic bacteria are spoilage organisms

Oxygen

- Aerobic bacteria must have oxygen to grow.

- Anaerobic bacteria can not grow when oxygen is present.

o Anaerobic bacteria grow well in vacuum packaged foods.

o Anaerobic bacteria also grow well in the middle of cooked food masses.

- Facultative anaerobic bacteria can grow with or without free oxygen.

o Most foodborne illness causing microorganisms are facultative

anaerobic.

- Microaerophilic organisms have very specific oxygen requirements, usually

around 3-6 percent.

Bacteria

Bacillus cereus

Campylobacter jejuni Parasites

Clostridium perfringens Anisakis spp.

Viruses

Clostridium botulinum Cryptosporidium parvum

Hepatitis A

Escherichia coli Cyclospora cayetonensis

Norwalk virus group

Listeria monocytogenes Giardia lamblia

Rotavirus

Salmonella spp. Toxoplasma gondii

Shigella spp. Trichinella spiralis

Staphylococcus aureus

Vibrio spp.

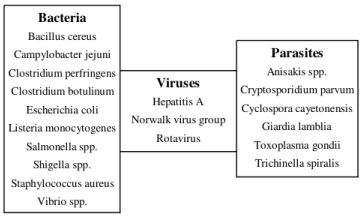
Return to Table of Contents

Temperature Conditions

- **Psychrophilic** bacteria grow within the range of 32°F to 70°F
  - Most are spoilage organisms, but some cause illness
- **Mesophilic** bacteria grow within the range of 70°F to 110°F
- **Thermophilic** bacteria grow above 110°F
  - All thermophilic bacteria are spoilage organisms

Oxygen

- **Aerobic** bacteria must have oxygen to grow.
- **Anaerobic** bacteria can not grow when oxygen is present.
  - **Anaerobic** bacteria grow well in vacuum packaged foods.
  - **Anaerobic** bacteria also grow well in the middle of cooked food masses.
- **Facultative anaerobic** bacteria can grow with or without free oxygen.
  - Most foodborne illness causing microorganisms are **facultative anaerobic.**
- **Microaerophilic** organisms have very specific oxygen requirements, usually around 3-6 percent.



Return to Table of Contents

Original source panel - page 3

Page 4 - Anisakiasis

Anisakiasis

Pathogen: Anisakis spp.

Type: Parasitic infection

Spore Former: No

Description: Anisakis spp. Are nematodes (roundworms) found in fish. They are usually

1 to 1 ½ inches long and the diameter of a human hair. They are beige, ivory, white, gray

brown, or pink.

Incubation Period: 1 hour to 2 weeks

Duration of Illness: less than 3 weeks

Symptoms:

-If the worm attaches to the throat: Coughing.

-If the worm attaches to the stomach: vomiting and abdominal pain.

-If the worm attaches to the large intestine: Sharp pain and fever, similar to appendicitis.

Common Cause: The worms are transferred to fish (mostly bottom feeders) through

seawater.

Foods Implicated: Raw, undercooked, or improperly frozen fish, including cod,

haddock, fluke, salmon, herring, flounder, monkfish. Sea urchins, crab, shrimp, starfish,

and tuna.

Prevention: Purchase seafood from approved dealers. Cook seafood to proper

temperatures. Properly freeze fish.

Return to Table of Contents

Symptoms: Vomit type: Nausea and vomiting, occasionally abdominal cramps and/or

diarrhea. Diarrhea type: Watery diarrhea, abdominal cramps and pain, nausea.

Common Cause: Most often associated with foods that are improperly stored (in the

“Danger Zone”), which allows spores to convert into vegetative cells. Vegetative cells

then produce toxin which leads to illness.

Foods Implicated: Vomit type: usually associated with grain products such as rice,

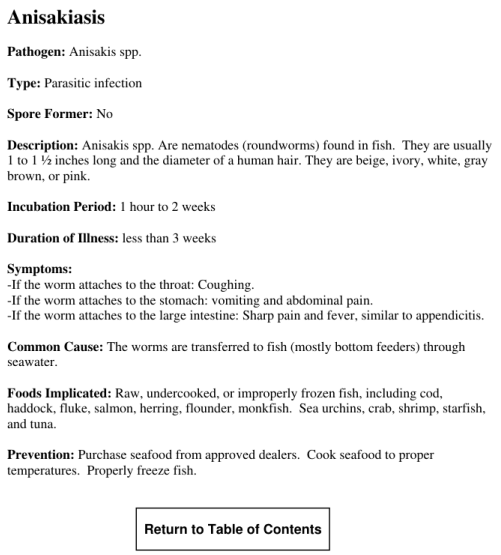
potatoes, pasta, corn, cornstarch, soybeans, tofu, and flour. Diarrhea type: meats, milk,

vegetables, and fish.

Prevention: Foods must be cooked and hot held properly and then cooled down quickly

within 4 hours.

Return to Table of Contents



Anisakiasis

Pathogen: Anisakis spp.

Type: Parasitic infection

Spore Former: No

Description: Anisakis spp. Are nematodes (roundworms) found in fish. They are usually 1 to 1 ½ inches long and the diameter of a human hair. They are beige, ivory, white, gray brown, or pink.

Incubation Period: 1 hour to 2 weeks

Duration of Illness: less than 3 weeks

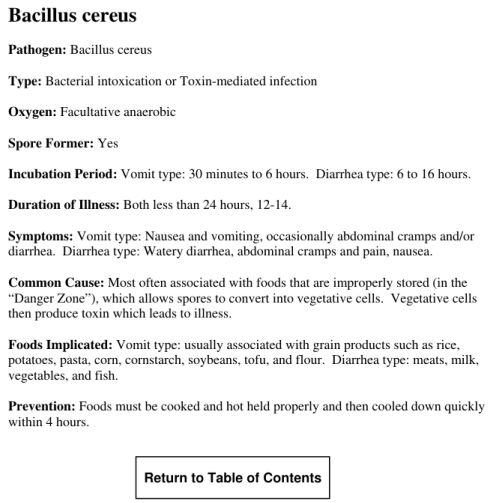
Symptoms:
-If the worm attaches to the throat: Coughing.
-If the worm attaches to the stomach: vomiting and abdominal pain.
-If the worm attaches to the large intestine: Sharp pain and fever, similar to appendicitis.

Common Cause: The worms are transferred to fish (mostly bottom feeders) through seawater.

Foods Implicated: Raw, undercooked, or improperly frozen fish, including cod, haddock, fluke, salmon, herring, flounder, monkfish. Sea urchins, crab, shrimp, starfish, and tuna.

Prevention: Purchase seafood from approved dealers. Cook seafood to proper temperatures. Properly freeze fish.

Return to Table of Contents



Bacillus cereus

Pathogen: Bacillus cereus

Type: Bacterial intoxication or Toxin-mediated infection

Oxygen: Facultative anaerobic

Spore Former: Yes

Incubation Period: Vomit type: 30 minutes to 6 hours. Diarrhea type: 6 to 16 hours.

Duration of Illness: Both less than 24 hours, 12-14.

Symptoms: Vomit type: Nausea and vomiting, occasionally abdominal cramps and/or diarrhea. Diarrhea type: Watery diarrhea, abdominal cramps and pain, nausea.

Common Cause: Most often associated with foods that are improperly stored (in the “Danger Zone”), which allows spores to convert into vegetative cells. Vegetative cells then produce toxin which leads to illness.

Foods Implicated: Vomit type: usually associated with grain products such as rice, potatoes, pasta, corn, cornstarch, soybeans, tofu, and flour. Diarrhea type: meats, milk, vegetables, and fish.

Prevention: Foods must be cooked and hot held properly and then cooled down quickly within 4 hours.

Return to Table of Contents

Original source panel - page 4

Page 5 - Bacillus cereus

Bacillus cereus

Pathogen: Bacillus cereus

Type: Bacterial intoxication or Toxin-mediated infection

Oxygen: Facultative anaerobic

Spore Former: Yes

Incubation Period: Vomit type: 30 minutes to 6 hours. Diarrhea type: 6 to 16 hours.

Duration of Illness: Both less than 24 hours, 12-14.

Original source panel - page 5

Page 6 - Botulism

Botulism

Pathogen: Clostridium botulinum

Type: Bacterial intoxication

Oxygen: Anaerobic

Spore Former: Yes

Incubation Period: 12 to 36 hours, but may vary from 4 hours to 8 days

Duration of Illness: Several days to a year

Symptoms: Dizziness, blurred or double vision, difficulty speaking, swallowing, and

breathing, fatigue, weakness, headache.

Common Cause: Improperly heat-processed foods, especially home-canned products.

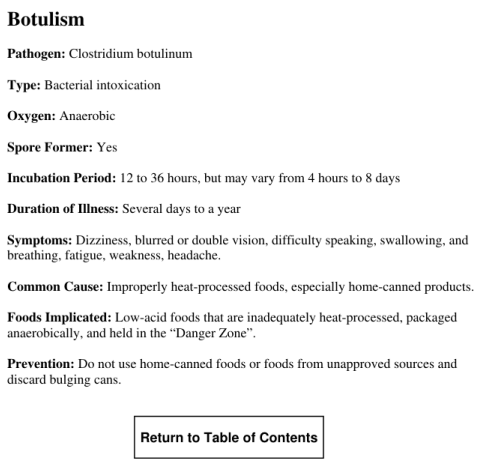
Foods Implicated: Low-acid foods that are inadequately heat-processed, packaged

anaerobically, and held in the “Danger Zone”.

Prevention: Do not use home-canned foods or foods from unapproved sources and

discard bulging cans.

Return to Table of Contents



Botulism

Pathogen: Clostridium botulinum

Type: Bacterial intoxication

Oxygen: Anaerobic

Spore Former: Yes

Incubation Period: 12 to 36 hours, but may vary from 4 hours to 8 days

Duration of Illness: Several days to a year

Symptoms: Dizziness, blurred or double vision, difficulty speaking, swallowing, and breathing, fatigue, weakness, headache.

Common Cause: Improperly heat-processed foods, especially home-canned products.

Foods Implicated: Low-acid foods that are inadequately heat-processed, packaged anaerobically, and held in the “Danger Zone”.

Prevention: Do not use home-canned foods or foods from unapproved sources and discard bulging cans.

Return to Table of Contents

Original source panel - page 6

Page 7 - Campylobacteriosis

Campylobacteriosis

Pathogen: Campylobacter jejuni

Type: Bacterial infection

Oxygen: Microaerophilic (3-6% oxygen)

Spore Former: No

Incubation Period: 2 to 5 days

Duration of Illness: 2 to 7 days, usually no more than 10 days (relapses common)

Symptoms: Diarrhea (watery and bloody), abdominal pain, fever, nausea and vomiting,

headache, muscle pain.

Common Cause: Often transferred from raw meats to other foods by cross

contamination, often via food contact surfaces such as a cutting board or food worker’s

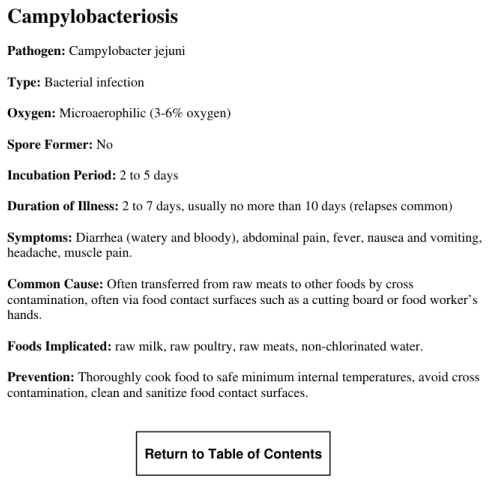
hands.

Foods Implicated: raw milk, raw poultry, raw meats, non-chlorinated water.

Prevention: Thoroughly cook food to safe minimum internal temperatures, avoid cross

contamination, clean and sanitize food contact surfaces.

Return to Table of Contents



Campylobacteriosis

Pathogen: Campylobacter jejuni

Type: Bacterial infection

Oxygen: Microaerophilic (3-6% oxygen)

Spore Former: No

Incubation Period: 2 to 5 days

Duration of Illness: 2 to 7 days, usually no more than 10 days (relapses common)

Symptoms: Diarrhea (watery and bloody), abdominal pain, fever, nausea and vomiting, headache, muscle pain.

Common Cause: Often transferred from raw meats to other foods by cross contamination, often via food contact surfaces such as a cutting board or food worker’s hands.

Foods Implicated: raw milk, raw poultry, raw meats, non-chlorinated water.

Prevention: Thoroughly cook food to safe minimum internal temperatures, avoid cross contamination, clean and sanitize food contact surfaces.

Return to Table of Contents

Original source panel - page 7

Page 8 - Ciguatera Fish Poisoning

Ciguatera Fish Poisoning

Pathogen: Ciguatoxin

Type: Biochemical intoxication

Oxygen:

Spore Former: No

Incubation Period: 15 minutes to 1 day

Duration of Illness: A few days, but may take several weeks or months

Symptoms: Dizziness, hot and cold flashes, diarrhea, vomiting, headache.

Common Cause: The toxin is transferred to finfish when they eat algae or other fish that

contain the toxin.

Foods Implicated: Fish, including barracuda, grouper, snapper, jack, mackerel, hogfish,

amberjack, blackjack, and triggerfish.

Prevention: Purchase seafood from approved dealers. Cooking will not destroy toxin.

Return to Table of Contents

Ciguatera Fish Poisoning

Pathogen: Ciguatoxin

Type: Biochemical intoxication

Oxygen:

Spore Former: No

Incubation Period: 15 minutes to 1 day

Duration of Illness: A few days, but may take several weeks or months

Symptoms: Dizziness, hot and cold flashes, diarrhea, vomiting, headache.

Common Cause: The toxin is transferred to finfish when they eat algae or other fish that contain the toxin.

Foods Implicated: Fish, including barracuda, grouper, snapper, jack, mackerel, hogfish, amberjack, blackjack, and triggerfish.

Prevention: Purchase seafood from approved dealers. Cooking will not destroy toxin.

Return to Table of Contents

Original source panel - page 8

Page 9 - Clostridium perfringens

Clostridium perfringens

Pathogen: Clostridium perfringens

Type: Toxin-mediated infection

Oxygen: Anaerobic

Spore Former: Yes

Incubation Period: 8 to 22 hours

Duration of Illness: 24 hours

Symptoms: Abdominal pain, diarrhea, nausea, dehydration, (fever and vomiting usually

absent).

Common Cause: Most often attributed to foods that are temperature abused, especially

those that have been improperly cooled and reheated.

Foods Implicated: Cooked meats and poultry that have been improperly cooled and

served cold or improperly reheated, stews, gravy, beans that have been cooled slowly.

Prevention: Foods must be cooled down quickly within 4 hours, reheated quickly to

165°F within 2 hours, and then held above 140°F.

Return to Table of Contents

Clostridium perfringens

Pathogen: Clostridium perfringens

Type: Toxin-mediated infection

Oxygen: Anaerobic

Spore Former: Yes

Incubation Period: 8 to 22 hours

Duration of Illness: 24 hours

Symptoms: Abdominal pain, diarrhea, nausea, dehydration, (fever and vomiting usually absent).

Common Cause: Most often attributed to foods that are temperature abused, especially those that have been improperly cooled and reheated.

Foods Implicated: Cooked meats and poultry that have been improperly cooled and served cold or improperly reheated, stews, gravy, beans that have been cooled slowly.

Prevention: Foods must be cooled down quickly within 4 hours, reheated quickly to 165°F within 2 hours, and then held above 140°F.

Return to Table of Contents

Original source panel - page 9

Page 10 - E. coli O157:H7 enteritis

E. coli O157:H7 enteritis

Pathogen: Escherichia coli O157:H7

Type: Bacterial infection or toxin-mediated infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 12 to 72 hours

Duration of Illness: 1 to 3 days

Symptoms: Serious threat for children up to age 16 and the elderly. Severe abdominal

pain, vomiting, bloody diarrhea, kidney failure and Hemolytic Uremic Syndrome.

Common Cause: Usually transferred to food by contact with intestines during slaughter.

Apples used for juice where cattle or deer also grazed. Infected employees that do not

properly wash their hands after going to the toilet.

Foods Implicated: undercooked beef and other red meats, improperly pasteurized milk,

raw finfish,

Prevention: Thoroughly cook ground meats to 165°F, make sure employees properly

wash their hands, Use only pasteurized apple cider, fruit juice, and milk products.

Return to Table of Contents

E. coli O157:H7 enteritis

Pathogen: Escherichia coli O157:H7

Type: Bacterial infection or toxin-mediated infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 12 to 72 hours

Duration of Illness: 1 to 3 days

Symptoms: Serious threat for children up to age 16 and the elderly. Severe abdominal pain, vomiting, bloody diarrhea, kidney failure and Hemolytic Uremic Syndrome.

Common Cause: Usually transferred to food by contact with intestines during slaughter. Apples used for juice where cattle or deer also grazed. Infected employees that do not properly wash their hands after going to the toilet.

Foods Implicated: undercooked beef and other red meats, improperly pasteurized milk, raw finfish,

Prevention: Thoroughly cook ground meats to 165°F, make sure employees properly wash their hands, Use only pasteurized apple cider, fruit juice, and milk products.

Return to Table of Contents

Return to Table of Contents

Food Allergens

Type of Illness: An allergic reaction usually involving the skin, mouth, digestive tract, or airways.

Symptoms:
-Skin: hives, rashes, and itching
-Mouth: swelling and itching of the lips and tongue
-Digestive tract: vomiting and diarrhea
-Airways: difficulty breathing, wheezing

Incubation Period: As little as 5 minutes

Common Cause:
Foods containing:
Milk
Egg
Wheat proteins
Peanuts
Tree nuts
Soy
Fish
Shellfish

Prevention: Avoid eating foods containing these ingredients, proper labeling of packaged and prepared foods.

Return to Table of Contents

Original source panel - page 10

Page 11 - Food Allergens

Food Allergens

Type of Illness: An allergic reaction usually involving the skin, mouth, digestive tract, or

airways.

Symptoms:

-Skin: hives, rashes, and itching

-Mouth: swelling and itching of the lips and tongue

-Digestive tract: vomiting and diarrhea

-Airways: difficulty breathing, wheezing

Incubation Period: As little as 5 minutes

Common Cause:

Foods containing:

Milk

Egg

Wheat proteins

Peanuts

Tree nuts

Soy

Fish

Shellfish

Prevention: Avoid eating foods containing these ingredients, proper labeling of

Original source panel - page 11

Page 12 - Hepatitis A

Hepatitis A

Pathogen: Hepatitis A virus

Type: Viral infection

Oxygen:

Spore Former: No

Incubation Period: 15 to 50 days, average 28-30 days

Duration of Illness: Several weeks, severe cases may last several months

Symptoms: Sudden onset of fever, nausea, vomiting, abdominal pain, fatigue, swelling

of liver, and jaundice.

Common Cause: Ingestion of food and water that contain the virus.

Foods Implicated: Raw and lightly cooked shellfish harvested from polluted waters.

Raw vegetables that have been washed with contaminated water. Most Potentially

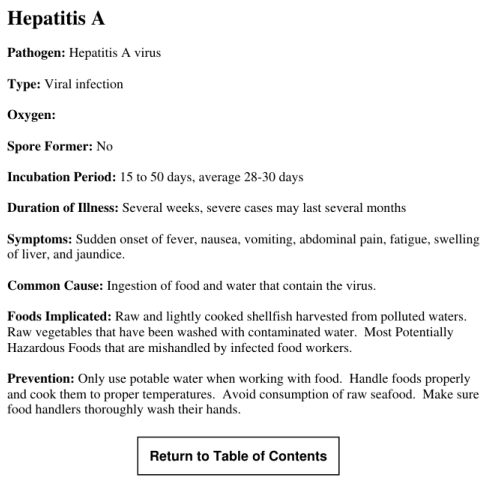
Hazardous Foods that are mishandled by infected food workers.

Prevention: Only use potable water when working with food. Handle foods properly

and cook them to proper temperatures. Avoid consumption of raw seafood. Make sure

food handlers thoroughly wash their hands.

Return to Table of Contents



Hepatitis A

Pathogen: Hepatitis A virus

Type: Viral infection

Oxygen:

Spore Former: No

Incubation Period: 15 to 50 days, average 28-30 days

Duration of Illness: Several weeks, severe cases may last several months

Symptoms: Sudden onset of fever, nausea, vomiting, abdominal pain, fatigue, swelling of liver, and jaundice.

Common Cause: Ingestion of food and water that contain the virus.

Foods Implicated: Raw and lightly cooked shellfish harvested from polluted waters. Raw vegetables that have been washed with contaminated water. Most Potentially Hazardous Foods that are mishandled by infected food workers.

Prevention: Only use potable water when working with food. Handle foods properly and cook them to proper temperatures. Avoid consumption of raw seafood. Make sure food handlers thoroughly wash their hands.

Return to Table of Contents

Original source panel - page 12

Page 13 - Listeriosis

Listeriosis

Pathogen: Listeria monocytogenes

Type: Bacterial infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 1 day to 3 weeks

Duration of Illness: Indefinite depending on treatment, can be life threatening for

pregnant and immuno-compromised.

Symptoms: Flu-like symptoms in healthy adults including nausea, vomiting, headache,

fever, chills, and backache. Can be life threatening (septicemia, meningitis, encephalitis,

and birth defects) for pregnant woman or immuno-compromised.

Common Cause: Cross contamination or undercooked foods.

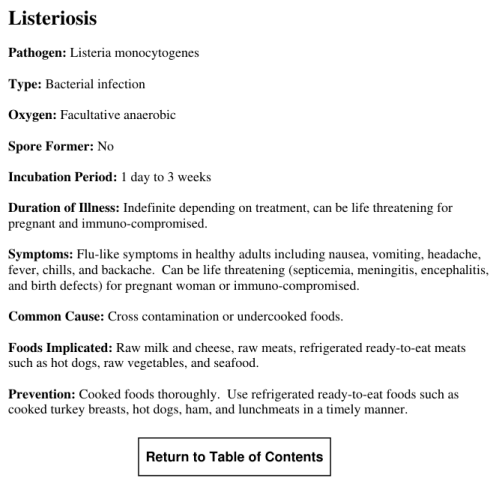
Foods Implicated: Raw milk and cheese, raw meats, refrigerated ready-to-eat meats

such as hot dogs, raw vegetables, and seafood.

Prevention: Cooked foods thoroughly. Use refrigerated ready-to-eat foods such as

cooked turkey breasts, hot dogs, ham, and lunchmeats in a timely manner.

Return to Table of Contents



Listeriosis

Pathogen: Listeria monocytogenes

Type: Bacterial infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 1 day to 3 weeks

Duration of Illness: Indefinite depending on treatment, can be life threatening for pregnant and immuno-compromised.

Symptoms: Flu-like symptoms in healthy adults including nausea, vomiting, headache, fever, chills, and backache. Can be life threatening (septicemia, meningitis, encephalitis, and birth defects) for pregnant woman or immuno-compromised.

Common Cause: Cross contamination or undercooked foods.

Foods Implicated: Raw milk and cheese, raw meats, refrigerated ready-to-eat meats such as hot dogs, raw vegetables, and seafood.

Prevention: Cooked foods thoroughly. Use refrigerated ready-to-eat foods such as cooked turkey breasts, hot dogs, ham, and lunchmeats in a timely manner.

Return to Table of Contents

Original source panel - page 13

Page 14 - Norovirus

Norovirus

Pathogen: caliciviruses (previously known as “Norwalk-like viruses”)

Type: Viral infection

Oxygen:

Spore Former: No

Incubation Period: Usually 1 to 2 days, as early as 12 hours

Duration of Illness: 1 to 2 days

Symptoms: Gastroenteritis (“stomach flu”) including nausea, vomiting, diarrhea,

abdominal pain, headache, low-grade fever, chills, muscle aches.

Common Cause: Ingestion of food and water that have been contaminated with feces

that contain the virus.

Foods Implicated: Raw and lightly cooked shellfish harvested from polluted waters.

Raw salad ingredients. Foods that are handled by infected food workers.

Prevention: Only use potable water when working with food. Handle foods properly

and cook them to proper temperatures. Avoid consumption of raw seafood. Make sure

food handlers thoroughly wash their hands.

Return to Table of Contents

Norovirus

Pathogen: caliciviruses (previously known as “Norwalk-like viruses”)

Type: Viral infection

Oxygen:

Spore Former: No

Incubation Period: Usually 1 to 2 days, as early as 12 hours

Duration of Illness: 1 to 2 days

Symptoms: Gastroenteritis (“stomach flu”) including nausea, vomiting, diarrhea, abdominal pain, headache, low-grade fever, chills, muscle aches.

Common Cause: Ingestion of food and water that have been contaminated with feces that contain the virus.

Foods Implicated: Raw and lightly cooked shellfish harvested from polluted waters. Raw salad ingredients. Foods that are handled by infected food workers.

Prevention: Only use potable water when working with food. Handle foods properly and cook them to proper temperatures. Avoid consumption of raw seafood. Make sure food handlers thoroughly wash their hands.

Return to Table of Contents

Salmonellosis

Pathogen: Salmonella spp.

Type: Bacterial infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 6 to 72 hours, usually 12 to 36 hours

Duration of Illness: 2 to 3 days

Symptoms: Abdominal pain, headache, nausea, vomiting, fever, diarrhea, dehydration.

Common Cause: Commonly through cross contamination when fecal material is transferred to food through contact with raw foods (especially poultry), contaminated food contact surfaces, or infected food workers.

Foods Implicated: Raw meat and poultry products, shell eggs, sliced fruits and vegetables, milk, dairy products, and other protein foods.

Prevention: Cook foods thoroughly. Clean and sanitize raw food contact surfaces after use. Make sure food workers wash their hands before working with food.

Return to Table of Contents

Original source panel - page 14

Page 15 - Salmonellosis

Salmonellosis

Pathogen: Salmonella spp.

Type: Bacterial infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 6 to 72 hours, usually 12 to 36 hours

Duration of Illness: 2 to 3 days

Symptoms: Abdominal pain, headache, nausea, vomiting, fever, diarrhea, dehydration.

Common Cause: Commonly through cross contamination when fecal material is

transferred to food through contact with raw foods (especially poultry), contaminated

food contact surfaces, or infected food workers.

Foods Implicated: Raw meat and poultry products, shell eggs, sliced fruits and

vegetables, milk, dairy products, and other protein foods.

Prevention: Cook foods thoroughly. Clean and sanitize raw food contact surfaces after

use. Make sure food workers wash their hands before working with food.

Return to Table of Contents

Original source panel - page 15

Page 16 - Scombroid Poisoning

Scombroid Poisoning

Pathogen: Scombrotoxin

Type: Biochemical intoxication

Oxygen:

Spore Former: No

Incubation Period: A few minutes to 30 minutes after consumption

Duration of Illness: 8 to 12 hours

Symptoms: Dizziness, burning or tingling sensation in mouth, facial rash or hives,

peppery taste in mouth, shortness of breath, headache, itchy, teary eyes.

Common Cause: Leaving fish in the Danger Zone.

Foods Implicated: Fish, including tuna, anchovies, blue fish, mackerel, amberjack,

abalone, and mahi-mahi. Swiss cheese.

Prevention: Purchase seafood from approved dealers. Keep “fresh” seafood out of the

Danger Zone. Do not accept seafood that is suspected of being thawed and refrozen or

temperature abused.

Return to Table of Contents

Scombroid Poisoning

Pathogen: Scombrotoxin

Type: Biochemical intoxication

Oxygen:

Spore Former: No

Incubation Period: A few minutes to 30 minutes after consumption

Duration of Illness: 8 to 12 hours

Symptoms: Dizziness, burning or tingling sensation in mouth, facial rash or hives, peppery taste in mouth, shortness of breath, headache, itchy, teary eyes.

Common Cause: Leaving fish in the Danger Zone.

Foods Implicated: Fish, including tuna, anchovies, blue fish, mackerel, amberjack, abalone, and mahi-mahi. Swiss cheese.

Prevention: Purchase seafood from approved dealers. Keep "fresh" seafood out of the Danger Zone. Do not accept seafood that is suspected of being thawed and refrozen or temperature abused.

Return to Table of Contents

Shigellosis

Pathogen: Shigella

Type: Bacterial infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 1 to 7 days, usually 1 to 3 days

Duration of Illness: 4 to 7 days, can be indefinite depending on when treatment is administered.

Symptoms: Diarrhea (sometimes bloody), fever, abdominal cramps, chills, fatigue, and dehydration.

Common Cause: Water that is contaminated by fecal material and food and utensils handled by infected food workers.

Foods Implicated: Common in ready-to-eat salads (i.e., potato, chicken), milk and dairy products, raw vegetables.

Prevention: Cook foods thoroughly, do not allow food workers that have been diagnosed with shigellosis to handle food, make sure food workers wash their hands before working with food, avoid cross contamination, and wash foods with potable water.

Return to Table of Contents

Original source panel - page 16

Page 17 - Shigellosis

Shigellosis

Pathogen: Shigella

Type: Bacterial infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 1 to 7 days, usually 1 to 3 days

Duration of Illness: 4 to 7 days, can be indefinite depending on when treatment is

administered.

Symptoms: Diarrhea (sometimes bloody), fever, abdominal cramps, chills, fatigue, and

dehydration.

Common Cause: Water that is contaminated by fecal material and food and utensils

handled by infected food workers.

Foods Implicated: Common in ready-to-eat salads (i.e., potato, chicken), milk and dairy

products, raw vegetables.

Prevention: Cook foods thoroughly, do not allow food workers that have been diagnosed

with shigellosis to handle food, make sure food workers wash their hands before working

with food, avoid cross contamination, and wash foods with potable water.

Return to Table of Contents

Original source panel - page 17

Page 18 - Staphylococcus

Staphylococcus

Pathogen: Staphylococcus aureus

Type: Bacterial intoxication

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 1 to 6 hours, usually 2 to 4 hours

Duration of Illness: 1 to 2 days

Symptoms: Nausea, vomiting, acute abdominal pains, diarrhea, changes in blood

pressure and heart rate.

Common Cause: Contamination from food worker's hands is the most common way the

organism is introduced into foods.

Foods Implicated: Cooked ready-to-eat foods, salads, meat, poultry, custards, high-salt

foods, milk and dairy products.

Prevention: Avoid contact of foods with bare hands, Make sure food workers wash their

hands and bandage and glove open wounds on their hands. Heat and cool foods properly.

Return to Table of Contents

Staphylococcus

Pathogen: Staphylococcus aureus

Type: Bacterial intoxication

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 1 to 6 hours, usually 2 to 4 hours

Duration of Illness: 1 to 2 days

Symptoms: Nausea, vomiting, acute abdominal pains, diarrhea, changes in blood pressure and heart rate.

Common Cause: Contamination from food worker's hands is the most common way the organism is introduced into foods.

Foods Implicated: Cooked ready-to-eat foods, salads, meat, poultry, custards, high-salt foods, milk and dairy products.

Prevention: Avoid contact of foods with bare hands, Make sure food workers wash their hands and bandage and glove open wounds on their hands. Heat and cool foods properly.

Return to Table of Contents

Original source panel - page 18

Vibrio spp.

Pathogen: Vibrio vulnificus, Vibrio parahaemolyticus, Vibrio cholera

Type: Bacterial infection

Oxygen:

Spore Former: No

Incubation Period: 2 hours to 2 days

Duration of Illness: 2 to 3 days, can last longer

Symptoms: Headache, fever, chills, diarrhea, vomiting, skin lesions, abdominal pain, septic shock.

Common Cause: Consumption of raw or lightly cooked seafood.

Foods Implicated: Fish and shellfish, including oysters, clams, mussels, crabs, shrimp, and lobster.

Prevention: Buy seafood, especially shellfish, from approved sources. Cook seafood and shellfish to proper temperatures and avoid eating raw or lightly cooked seafood and shellfish. Avoid cross contamination.

Person's experiencing any of these symptoms after eating these foods should seek medical attention immediately.

Return to Table of Contents

Original source panel - page 19

Page 19 - Vibrio spp.

Vibrio spp.

Pathogen: Vibrio vulnificus, Vibrio parahaemolyticus, Vibrio cholera

Type: Bacterial infection

Oxygen:

Spore Former: No

Incubation Period: 2 hours to 2 days

Duration of Illness: 2 to 3 days, can last longer

Symptoms: Headache, fever, chills, diarrhea, vomiting, skin lesions, abdominal pain,

septic shock.

Common Cause: Consumption of raw or lightly cooked seafood.

Foods Implicated: Fish and shellfish, including oysters, clams, mussels, crabs, shrimp,

and lobster.

Prevention: Buy seafood, especially shellfish, from approved sources. Cook seafood

and shellfish to proper temperatures and avoid eating raw or lightly cooked seafood and

shellfish. Avoid cross contamination.

Person's experiencing any of these symptoms after eating these foods should seek

medical attention immediately.

Return to Table of Contents